

AP Precalc Unit 2 Wkst B Solutions

(graphs created during IF by student E.H.) editor's note: really good examples of why you need graph paper for graphs)

1) $y = \csc \pi x$

amp = 1

midline: $y = 0$

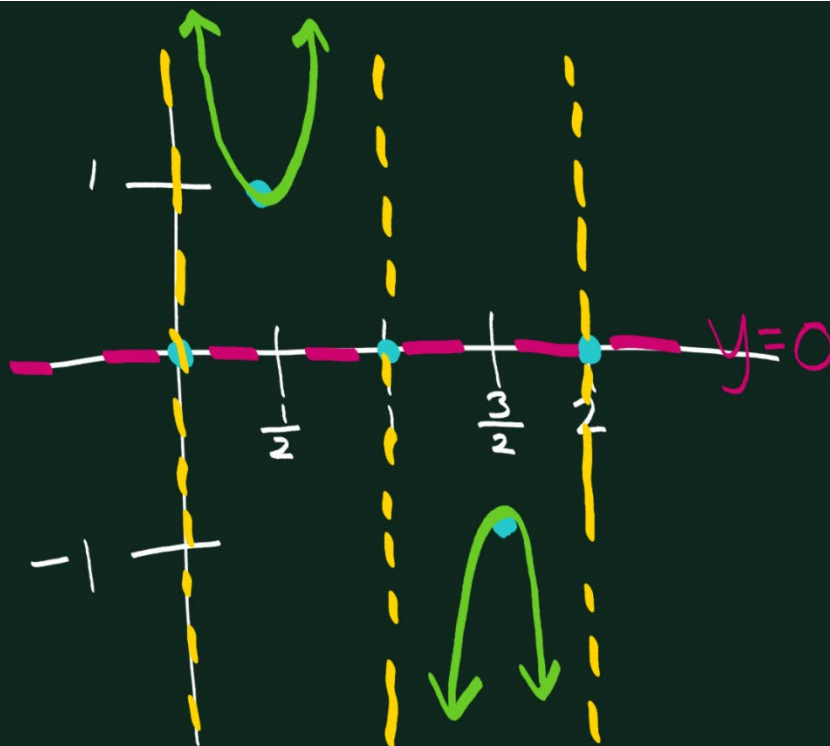
LOP = 2

boundaries: $\pi x = 0$ $\pi x = 2\pi$

$2 \cdot \frac{1}{4} = \frac{1}{2} \rightarrow \perp$ $x = 0$ $x = 2$

5 x-values: $0, \frac{1}{2}, 1, \frac{3}{2}, 2$

Sine-



2) $y = \sec \pi x - 1$

amp = 1

midline: $y = -1$

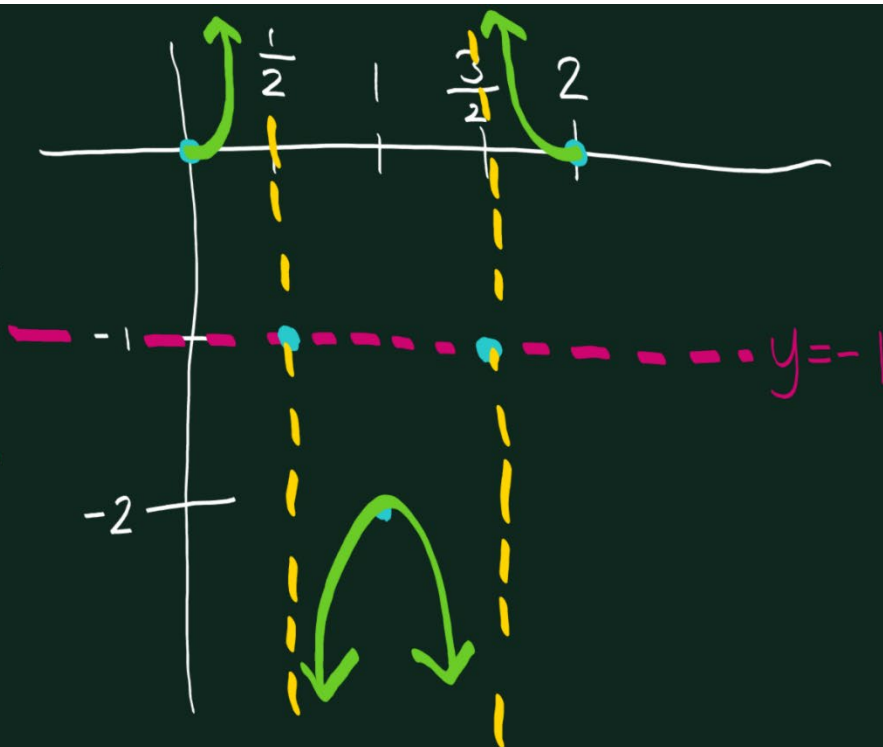
LOP = 2

boundaries: $\pi x = 0$ $\pi x = 2\pi$

$2 \cdot \frac{1}{4} = \frac{1}{2} \rightarrow \perp$ $x = 0$ $x = 2$

5 x-values: $0, \frac{1}{2}, 1, \frac{3}{2}, 2$

cosine =



3) $y = -2\sec 4x + 2$

amp = 2

midline: $y = 2$

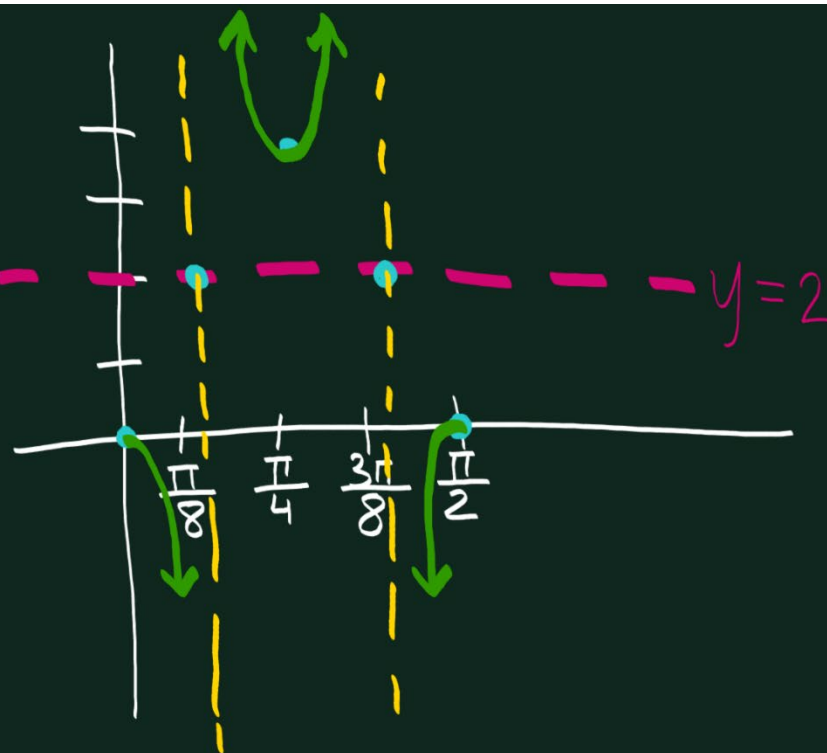
LOP = $\frac{\pi}{2}$

boundaries: $4x = 0$ $4x = 2\pi$
 $x = 0$ $x = \frac{\pi}{2}$

$\frac{\pi}{2} \cdot \frac{1}{4} = \frac{\pi}{8}$

5 x-values: $0, \frac{\pi}{8}, \frac{\pi}{4}, \frac{3\pi}{8}, \frac{\pi}{2}$

- secant



4) $y = -3\csc(\pi x + 3\pi) - 4$

amp = 3

midline: $y = -4$

LOP = $\frac{\pi}{2}$

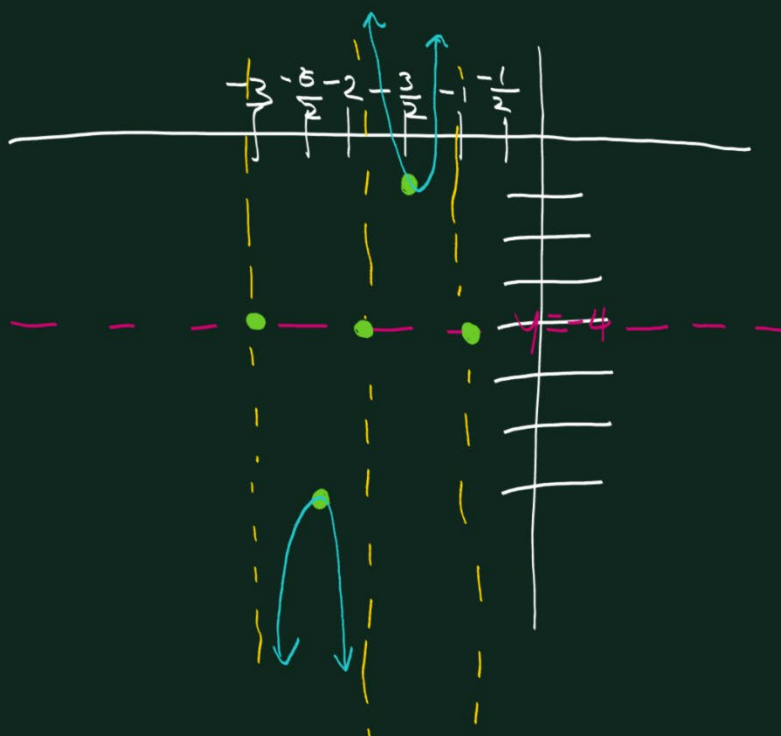
boundaries: $\pi x + 3\pi = 0$ $\pi x + 3\pi = 2\pi$

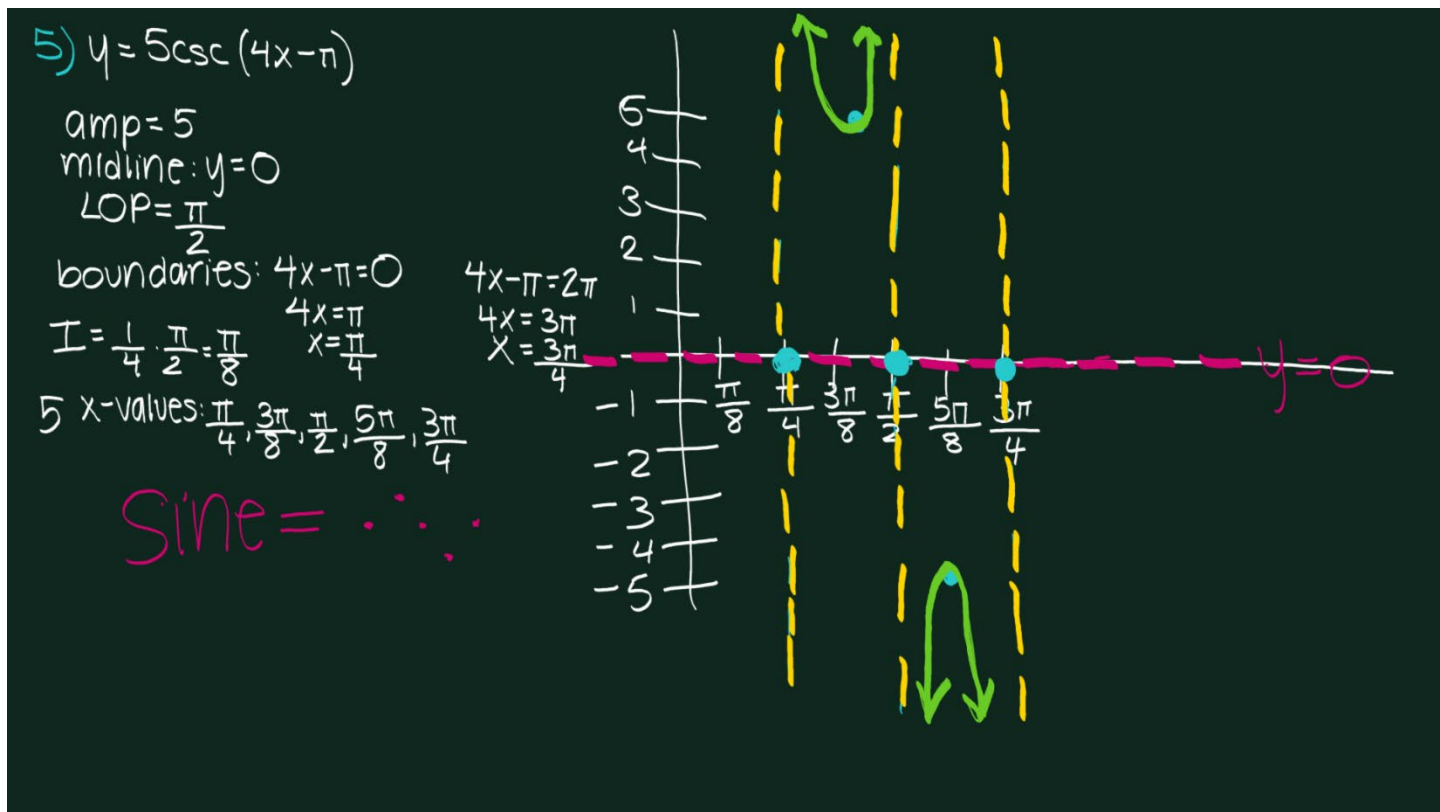
$2 \cdot \frac{1}{4} = \frac{1}{2}$ $\pi x = -3\pi$
 $x = -3$

$\pi x = -\pi$
 $x = -1$

$-3, -\frac{5}{2}, -2, -\frac{3}{2}, -1$

sine





- Horizontal dilation by a factor of $\frac{1}{\pi}$ (horizontal shrink)
 Domain: $(-\infty, \infty), x \neq n$ Range: $(-\infty, -1] \cup [1, \infty)$
 Interval where function values are increasing, while the rate of change is decreasing: $(1, \frac{3}{2})$
- Horizontal dilation by a factor of $\frac{1}{\pi}$ (horizontal shrink) and a vertical translation (shift) down 1 unit.
 Domain: $(-\infty, \infty), x \neq \frac{1}{2} + n$ Range: $(-\infty, -2] \cup [0, \infty)$
 Interval where function values are increasing, while the rate of change is decreasing: $(\frac{1}{2}, 1)$
- Reflection over the midline, vertical dilation by a factor of 2 (vertical stretch), horizontal dilation by a factor of $\frac{1}{4}$ (horizontal shrink), and a vertical translation (shift) up 2 units.
 Domain: $(-\infty, \infty), x \neq \frac{\pi}{8} + n\frac{\pi}{4}$ Range: $(-\infty, 0] \cup [4, \infty)$
 Interval where function values are increasing, while the rate of change is decreasing: $(\frac{3\pi}{8}, \frac{\pi}{2})$
- Reflection over the midline, horizontal dilation by a factor of $\frac{1}{\pi}$ (horizontal shrink), horizontal translation (shift) 3 units to the left, and a vertical translation (shift) down 4 units.
 Domain: $(-\infty, \infty), x \neq n$ Range: $(-\infty, -7] \cup [-1, \infty)$
 Interval where function values are increasing, while the rate of change is decreasing: $(-3, -\frac{5}{2})$